CAPSTONE PROJECT EVALUATION • CSE (AI/ML)

Presento

AI-Powered Smart Attendance & Analytics System

SMART PRESENCE, VERIFIED.

Department of Computer Science & Engineering (AI & ML)

PROJECT TEAM PRN :

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INSTITUTIONAL METADATA

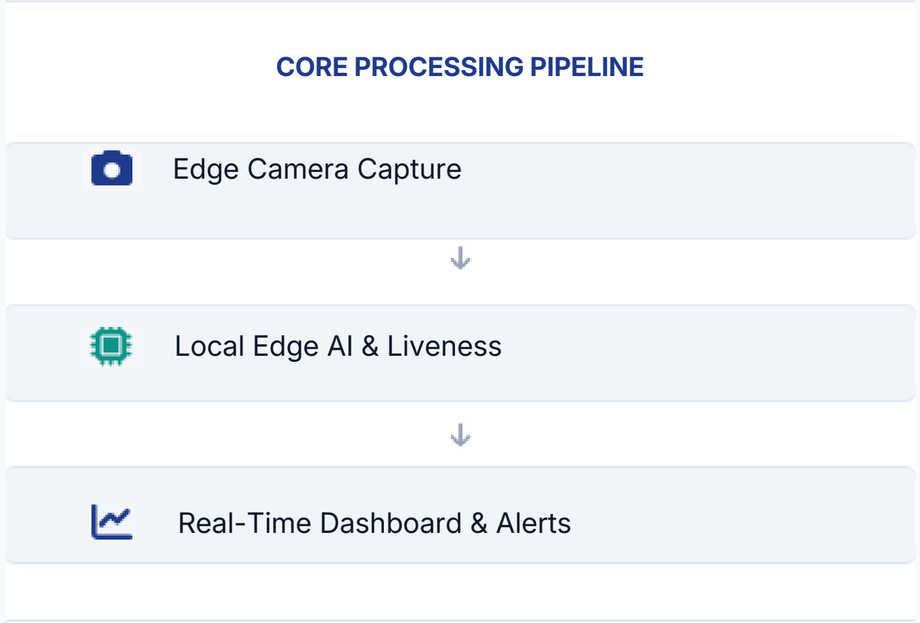
College: D.K.T.E College Of Engineering , Ichalkaranji.

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System Introduction

Presento is an IoT-based edge intelligence platform that completely automates classroom presence tracking through real-time facial recognition.

- **Edge Hardware Deployment:** Fully integrated on Raspberry Pi edge devices, executing real-time detection without latency or cloud dependence.
- **End-to-End Synergy:** Unifies computer vision algorithms, anti-spoofing machine learning, and a modern web management dashboard.
- **Institutional Modernization:** Replaces manual roll-calls and prone biometric hardware with a contactless, fraud-resistant solution.



Problem Statement

Manual Attendance

Manual roll-call consumes 5–10 minutes per lecture period. Over an academic semester, this accumulates to dozens of lost teaching hours across departments.

Proxy Attendance Fraud

Traditional sign-in sheets and basic biometric systems are easily bypassed through buddy-signing or fraudulent proxy checking, compromising record integrity.

Late Risk Detection

Students with low or declining attendance may not be identified early enough for intervention.

Lack of Structured Analytics

Traditional systems only show present or absent status and do not provide useful insights into late arrivals, absence patterns, or attendance risks.

Proposed Solution

Edge Facial Recognition

Autonomous recognition running on Raspberry Pi hardware mounted at classroom entry points, handling simultaneous multi-face detection in real time.

Liveness Verification

Integrated blink-detection algorithms (Eye Aspect Ratio) prevent spoofing attempts via static photographs, video replays, or digital screens.

Predictive Web Dashboard

Instructors access live attendance tracking, trend analysis, and automated predictive indicators flagging at-risk attendance thresholds.

Proactive Notification System

Automated Twilio/WhatsApp triggers instantly alert students and guardians regarding unexcused absences or impending attendance shortages.

IoT Hardware Architecture

Raspberry Pi 4 (4GB/8GB):
Core embedded computer executing localized CV algorithms and database sync.

Pi Camera / USB Webcam:
High-definition video sensor capturing classroom entry feeds at 30 FPS.

PIR Motion Sensor:
Triggers camera capture upon physical presence, optimizing idle power.

Audio Speaker / Buzzer:
Emits local audible chime confirming successful identity verification.

Power Unit (5V / 3A USB-C):
Ensures sustained industrial-grade power delivery for continuous edge execution.



AI & Machine Learning Framework

MediaPipe Framework
Performs lightweight, real-time multi-face detection and extracts 468 3D facial landmark meshes.

Face Recognition (OpenCV)
Generates 128-dimensional facial vector embeddings and calculates Euclidean distance matches.

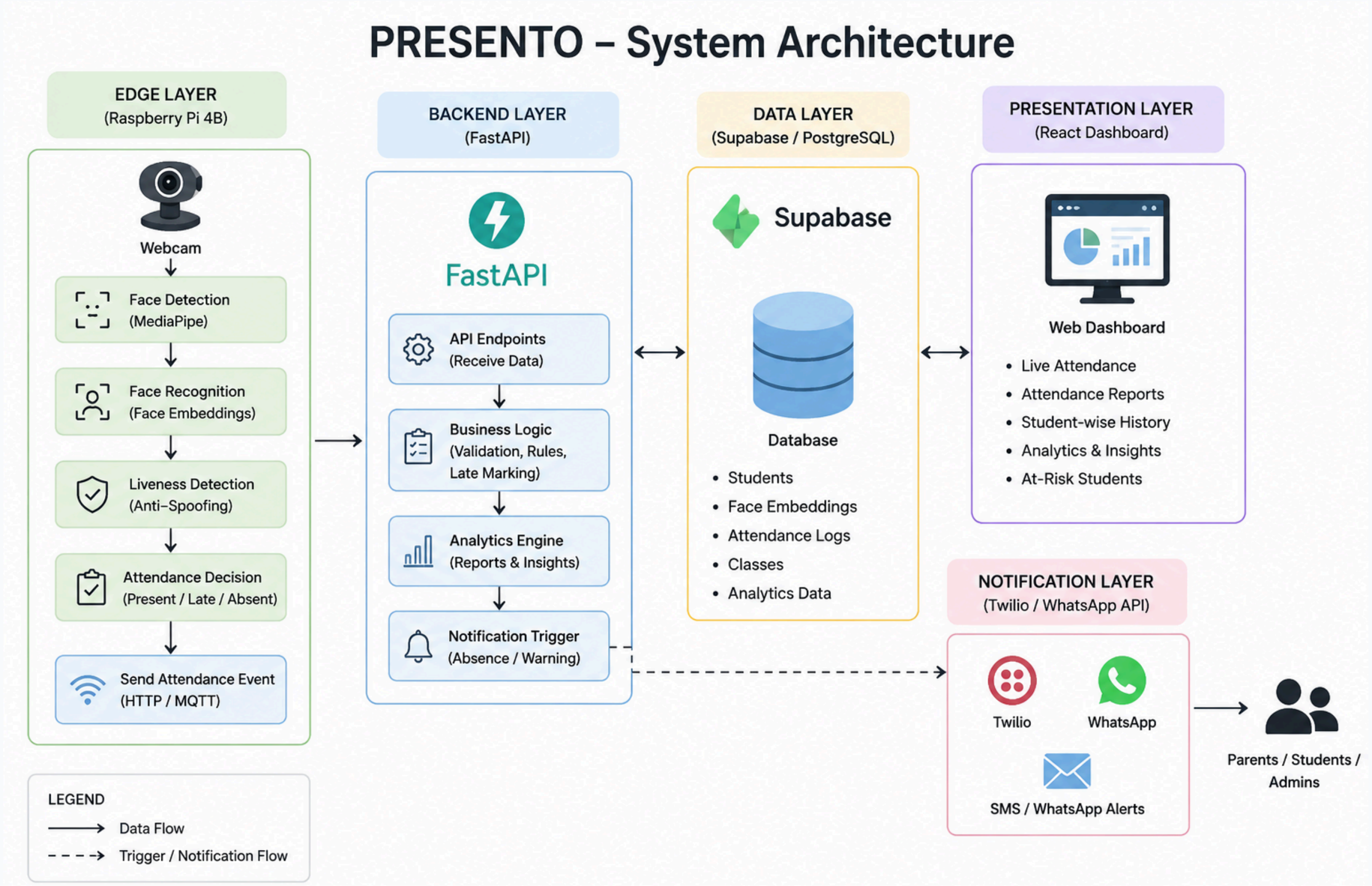
EAR Liveness Detection
Computes Eye Aspect Ratio (EAR) across consecutive frames to verify natural human blinking patterns.

Scikit-Learn Predictive Model
Trains supervised classification models on historical attendance trends to forecast at-risk trajectories.

MACHINE LEARNING EXECUTION PIPELINE

Raw Video Feed › Landmarks › 128-D Embeddings › Liveness EAR Check › Verified Attendance

System Architecture



Methodology & Operational Steps

1. **Student Enrollment:** Multi-angle face images captured, converted to 128-D vector embeddings, and registered in Supabase.
2. **Continuous Sensing:** Raspberry Pi camera scans classroom entrance upon motion detection by PIR sensor.
3. **Identity Match & Liveness:** Local embeddings matched against database; EAR blink ratio validates live human presence.
4. **Attendance Logging:** Verified record stamped with exact timestamp; late arrivals auto-tagged based on timetable logic.
5. **Backend Sync & Analytics:** Event pushed via FastAPI endpoint; scikit-learn model updates student risk score.
6. **Dashboard & Alerts:** React UI displays real-time class analytics automated Twilio SMS triggers for unexcused absences.

System Advantages

Time Efficiency
Eliminates 5-10 minutes of manual attendance taking per lecture, maximizing core teaching time.

Fraud Elimination
Anti-spoofing EAR liveness detection prevents proxy attendance via photos or screen recordings.

Real-Time Visibility
Provides faculty and administration with instant, live classroom presence status and dashboards.

Predictive Alerts
Machine learning analytics proactively identify students trending toward mandatory attendance shortfalls.

Cost-Effective
Built on off-the-shelf single-board computers (Raspberry Pi), enabling campus-wide scalability.

Privacy Preserving
Raw video frames are processed in RAM on the edge and discarded; only vector embeddings are stored.

System Limitations & Technical Constraints

Environmental Lighting Sensitivity

The system may have difficulty detecting faces properly in very dark areas or when strong light comes from behind the person.

Embedding Search Scaling

As the number of registered students increases, searching for the correct face can become slower on the Raspberry Pi.

Network Dependency for Sync

Face recognition works on the Raspberry Pi, but sending attendance data to the database and notifications requires a stable internet connection.

Identical Twin Ambiguity

The system may have difficulty correctly identifying identical twins because their facial features are very similar.

Future Research & Development Scope

Native Mobile Application

A dedicated mobile app can allow students and parents to check attendance, view updates, and request leave from their phones.

Institutional ERP & SIS Integration

Presento can be connected with college or university systems to automatically share attendance records and reduce manual data entry.

Multi-Camera Edge Mesh

Multiple cameras can be placed at different entrances or classrooms and connected together to track students from different locations.

3D Depth-Sensing Liveness

Adding depth sensors can make liveness detection more accurate and provide stronger protection against photos, videos, or 3D face masks.

Conclusion

Presento provides a smart and efficient solution for automating student attendance using AI, computer vision, and IoT. The system uses a Raspberry Pi and camera for real-time face recognition and liveness detection, helping reduce manual work and prevent proxy attendance.

References & Academic Citations

Ref #	Source Title & Author / Platform	Category
[1]	**Google AI Edge:** MediaPipe Face Mesh & Landmark Detection Framework Documentation (2024).	FRAMEWORK
[2]	**Geitgey, A.:** *face_recognition: The world's simplest facial recognition api for Python* (dlib-based C++ library).	ALGORITHM
[3]	**ResearchGate**"Automatic Class Attendance System Using Biometric Facial Recognition on Raspberry Pi Microprocessor."	RESEARCH
[4]	**FastAPI Framework:** Asynchronous Web Framework for Python 3.8+ Official Documentation.	BACKEND
[5]	**Soukupová, T., & Cech, J.:** "Real-Time Eye Blink Detection Using Facial Landmarks." *Computer Vision winter workshop* (2016).	RESEARCH
[6]	**Raspberry Pi Foundation:** Raspberry Pi 4 Model B Hardware & Camera Module Specification Manuals.	HARDWARE